

# Ngoc-Son Nguyen

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## RESEARCH INTERESTS

Multimodal AI, Generative AI, Speech Processing, Audio-Visual Speech Generation, Visual Voice Cloning, Video-to-Speech Synthesis

## EDUCATION

University of Science - Vietnam National University HCMC (HCMUS-VNU), Vietnam

Sep 2020 - Sep 2024

Bachelor of Science in Data Science

Graduated with **Highest Distinction** (GPA: 9.27/10.00, 3.86/4.00 – Ranked **2nd** in the Class of 2024)

**Thesis:** Advancing Vietnamese Visual Question Answering with Transformer and Convolutional Integration (Score: **10.0/10.0**)

**Thesis Supervisor:** Dr. Tung Le

## PUBLICATIONS & PREPRINTS

(\*): Equal contribution

- [P1] **Ngoc-Son Nguyen**, Thanh V. T. Tran, Jeongsoo Choi, Hieu-Nghia Huynh-Nguyen, Truong-Son Hy, Van Nguyen. [DiFlowDubber: Discrete Flow Matching for Automated Video Dubbing via Cross-Modal Alignment and Synchronization](#), *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPR) Findings, 2026*
- [P2] **Ngoc-Son Nguyen**, Thanh V. T. Tran, Hieu-Nghia Huynh-Nguyen, Truong-Son Hy, Van Nguyen. [DiFlow-TTS: Compact and Low-Latency Zero-Shot Text-to-Speech with Discrete Flow Matching](#), *Interspeech (Long Paper track), 2026*
- [P3] Thanh V. T. Tran, **Ngoc-Son Nguyen**, Luong Tran, Long-Khanh Pham, Paarth Neekhara, Shehzeen Samarah Hussain, Van Nguyen. [Precise Video-to-Audio Generation with Cross-Modal Alignment in Latent Space](#), (*Under Review*)
- [P4] Hieu-Nghia Huynh-Nguyen, Huynh Nguyen Dang, **Ngoc-Son Nguyen**, Van Nguyen. [Flamed-TTS: Flow Matching Attention-Free Models for Efficient Generating and Dynamic Pacing Zero-shot Text-to-Speech](#), (*Under Review*)
- [P5] Cuong Tran Van\*, Trong-Thang Pham\*, **Ngoc-Son Nguyen**, Duy Minh Ho Nguyen, Ngan Le. [DuFal: Dual-Frequency-Aware Learning for High-Fidelity Extremely Sparse-view CBCT Reconstruction](#), *Transactions on Machine Learning Research (TMLR 2025)*
- [P6] Trong-Thang Pham, AKASH AWASTHI, Saba Khan, Esteban Duran Marti, Tien-Phat Nguyen, Khoa Vo, Minh Tran, **Ngoc-Son Nguyen**, Cuong Tran Van, Yuki Ikebe, Anh Totti Nguyen, Anh Nguyen, Zhigang Deng, Carol C. Wu, Hien Van Nguyen, Ngan Le. [CT-ScanGaze: A Dataset and Baselines for 3D Volumetric Scanpath Modeling](#), *Proceedings of the IEEE/CVF International Conference on Computer Vision 2025 (ICCV 2025)* (**Highlight**)
- [P7] Hieu-Nghia Huynh-Nguyen, **Ngoc-Son Nguyen**, Huynh Nguyen Dang, Thieu Vo, Truong-Son Hy, Van Nguyen. [OZSpeech: One-step Zero-shot Speech Synthesis with Learned-Prior-Conditioned Flow Matching](#), *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*
- [P8] Khai Le-Duc\*, Ryan Zhang\*, **Ngoc-Son Nguyen\***, Tan-Hanh Pham, Anh Dao, Ba Hung Ngo, Anh Totti Nguyen, Truong-Son Hy. [LiteGPT: Large Vision-Language Model for Joint Chest X-ray Localization and Classification Task](#), (*Preprint*)
- [P9] **Ngoc-Son Nguyen**, Van Son Nguyen, Tung Le. [Advancing Vietnamese Visual Question Answering with Transformer and Convolutional Integration](#), *Computers and Electrical Engineering (Q1, IF = 4.9)*

## RESEARCH EXPERIENCE

AI Center, FPT Software

Ho Chi Minh, Vietnam

AI Research Resident

Aug 2024 - Present

**Supervisor:** Dr. Van Nguyen, Prof. Truong-Son Hy, Prof. Ngan Le

**Research topics:** Multimodal AI, Generative AI, Speech Processing, Medical Imaging

- Proposed a novel vision-language model tailored for medical imaging, specifically for joint localization and classification tasks on chest x-rays. The unified framework integrates multiple pre-trained visual encoders to enhance diagnostic accuracy. Established a strong baseline for future research, conducted ablation studies, and performed comprehensive result analysis. [P8] (*arXiv 2024*)
- Proposed an idea of using a learned prior as initialization instead of starting from Gaussian noise, followed by a rectified flow transformation to reach the target distribution, demonstrating improved effectiveness. Implemented the Montreal Forced Aligner to align phonemes with audio, and conducted experiments that yielded valuable insights into emerging challenges. [P7] (*ACL 2025*)
- Conducted extensive experiments to demonstrate the advantages of the proposed dataset for 3D volumetric scanpath modeling. Proposed a dual-encoding architecture with a Frequency-Enhanced Dual-Encoding design, which processes X-ray projections through parallel Spatial and Frequency encoders. This approach mitigates the inherent low-frequency bias of conventional CNN-based methods, enabling the model to simultaneously capture global structural context and fine-grained high-frequency anatomical details (e.g., pulmonary nodules and mandibular canals) that are critical for medical imaging. [P6] (*ICCV 2025*), [P5] (*TMLR 2025*)

- Conducted extensive benchmarking experiments comparing the proposed method with state-of-the-art baselines, including effective prompt design for SoundCap dataset construction, data preprocessing, and comprehensive evaluation to assess performance and robustness. [P4] (*arXiv 2025*), [P3] (*Under Review*)
- Proposed DiFlow-TTS, a novel zero-shot text-to-speech (TTS) system that investigates discrete flow matching as a viable direction for generative speech modeling. This work serves as an entry point for further research in discrete flow-based speech generation. Extensive empirical evaluations analyze both the strengths and limitations of the approach across key aspects, including naturalness, expressive attributes, speaker identity, and inference latency. [P2] (**Interspeech 2026**)
- Proposed DiFlowDubber, a video dubbing framework with a two-stage training pipeline that effectively transfers knowledge from a pre-trained text-to-speech (TTS) model to video-driven dubbing. The method employs a discrete flow matching generative backbone to address common limitations of TTS-based adaptation pipelines, including weak expressive prosody, limited acoustic richness, and imprecise audio-visual synchronization. [P1] (**CVPR 2026 Findings**)

**Multi-modal Machine Comprehension Group, FIT-HCMUS**

*Research Assistant*

Ho Chi Minh, Vietnam

Sep 2023 - Jun 2024

**Research topics:** Vietnamese Visual Question Answering

- Proposed the integration of Transformer architectures with Convolutional Neural Networks to improve image information extraction. Studied effective fusion techniques for visual and textual information, and conducted extensive research to optimize model performance on the ViVQA dataset, achieving state-of-the-art results. [P9] (*Journal Computers and Electrical Engineering 2024*)

## HONORS & AWARDS

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|-----------------------------------------------------------------------------------------------------------|---------------------|
| Outstanding Graduate Award, Vietnam National University, Ho Chi Minh City (VNU-HCMC)                      | Dec 2024            |
| Outstanding Graduate in Data Science Award, University of Science, VNU-HCMC (HCMUS)                       | Dec 2024            |
| Outstanding Achievement Award in Science and Technology Research, University of Science, VNU-HCMC (HCMUS) | Nov 2024            |
| University Scholarship for Excellent Academic Achievement, University of Science, VNU-HCMC (HCMUS)        | Sep 2020 - Sep 2023 |

## TECHNICAL SKILLS

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- **Programming:** C/C++, Python.
- **Framework:** PyTorch, Pytorch Lightning, SQL, Scikit-learn, Pandas, Matplotlib.
- **Tools:** Linux, Git/GitHub, L<sup>A</sup>T<sub>E</sub>X.
- **Languages:** Vietnamese (Native), English (740/990 TOEIC Listening and Reading; 310/400 TOEIC Speaking and Writing).

## REFERENCES

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**Dr. Van Nguyen**

*Principal AI Scientist / Tech Lead*

*UnitedHealth Group, United States*

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Scholar Profiles: Google Scholar

**Prof. Truong-Son Hy**

*Tenure-Track Assistant Professor*

*Department of Computer Science & Heersink School of Medicine*

*The University of Alabama at Birmingham, United States*

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Scholar Profiles: Personal Page — Google Scholar

**Prof. Ngan Le**

*Assistant Professor, Director of Artificial Intelligence and Computer Vision Lab*

*Department of CSCE*

*University of Arkansas, United States*

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**Dr. Tung Le**

*Lecturer*

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